

UFUG2106 Discrete Mathematics

Assignment 2

March 16, 2026

Q1 Solution

Note: The domain for all variables is $\mathbb{Z}^+$ (the set of positive integers: $1, 2, 3, 4, \dots$).

(a) $\exists x \sim Odd(x)$

- **Translation:** There exists a positive integer that is not odd.
- **Truth Value:** True
- **Explanation:** Positive integers like 2, 4, or 6 exist in the domain and are not odd.

(b) $\forall x (Prime(x) \rightarrow Greater(x, 1))$

- **Translation:** For every positive integer, if it is prime, then it is greater than 1.
- **Truth Value:** True
- **Explanation:** By definition, a prime number is an integer strictly greater than 1 that has no positive divisors other than 1 and itself. The smallest prime number is 2, which is greater than 1.

(c) $\exists x \forall y (Greater(y, x))$

- **Translation:** There is a positive integer that is strictly less than all positive integers.
- **Truth Value:** False
- **Explanation:** This statement claims there is an x such that every y in the domain is strictly greater than x . Because the domain includes x itself, this would mean x must be strictly greater than x ($x > x$), which is a logical contradiction.

(d) $\forall x (Even(x) \rightarrow \exists y (Equal(x, y + 2)))$

- **Translation:** For every even positive integer, there is a positive integer that is exactly 2 less than it.
- **Truth Value:** False
- **Explanation:** The domain is strictly positive integers ($\mathbb{Z}^+$), meaning 0 is not included. The number 2 is an even positive integer. However, there is no positive integer y that satisfies $2 = y + 2$, because that would require $y = 0$, which is outside the domain.

(e) $\exists x \exists y (Prime(x) \wedge Prime(y) \wedge Equal(x + y, 10))$

- **Translation:** There exist two prime positive integers whose sum equals 10.
- **Truth Value:** True
- **Explanation:** We can find positive integers in the domain that satisfy this. For example, $x = 3$ and $y = 7$ are both prime, and $3 + 7 = 10$. (Alternatively, $x = 5$ and $y = 5$ also works.)

Q2 Solution

(a) $\exists x (Even(x) \wedge \neg Prime(x))$

(b) $\exists x \exists y (Odd(x) \wedge Odd(y) \wedge \neg Equal(x, y) \wedge Even(x + y))$

(c) $\forall x \exists y (Greater(y, x))$

(d) $\exists x (Even(x) \wedge Prime(x) \wedge \forall y ((Even(y) \wedge Prime(y)) \rightarrow Equal(x, y)))$

(e) $\forall x (Prime(x) \rightarrow \exists y (Prime(y) \wedge Greater(y, x)))$

Q3 Solution

(a) $\exists x (P(x) \wedge Q(x)) \wedge \forall x R(x)$

- **Quantifier $\exists x$:**
 - Scope: $(P(x) \wedge Q(x))$
 - Bound variable: x (the occurrences inside its scope)
- **Quantifier $\forall x$:**
 - Scope: $R(x)$
 - Bound variable: x (the occurrence inside $R(x)$)
- **Free variables:** None. All occurrences of variables are bound by a quantifier.

(b) $\forall x (P(x) \rightarrow Q(x, y))$

- **Quantifier $\forall x$:**
 - Scope: $(P(x) \rightarrow Q(x, y))$
 - Bound variable: x
- **Free variables:** y . It is not bound by any quantifier in the formula.

(c) $\exists x (P(x) \wedge \forall y R(x, y))$

- **Quantifier $\exists x$:**
 - Scope: $(P(x) \wedge \forall y R(x, y))$
 - Bound variable: x
- **Quantifier $\forall y$:**
 - Scope: $R(x, y)$
 - Bound variable: y
- **Free variables:** None. Both x and y are properly bound within their respective scopes.

(d) $\forall x (P(x) \rightarrow Q(y)) \rightarrow \exists y (R(x) \wedge L(x, y, z))$

- **Quantifier $\forall x$:**
 - Scope: $(P(x) \rightarrow Q(y))$
 - Bound variable: x (only the occurrence in $P(x)$)
- **Quantifier $\exists y$:**
 - Scope: $(R(x) \wedge L(x, y, z))$
 - Bound variable: y (only the occurrence in $L(x, y, z)$)
- **Free variables:** x, y, z .
 - Note: The x in the second half ($R(x)$ and $L(x, y, z)$) is free because it falls outside the scope of $\forall x$. The y in the first half ($Q(y)$) is free because it falls outside the scope of $\exists y$. The variable z is entirely unbound.

(e) $\forall x P(x, y, z) \vee (\exists u Q(x, u) \rightarrow \exists w Q(y, w))$

- **Quantifier $\forall x$:**
 - Scope: $P(x, y, z)$
 - Bound variable: x (in $P(x, y, z)$)

- **Quantifier $\exists u$:**
 - Scope: $Q(x, u)$
 - Bound variable: u
- **Quantifier $\exists w$:**
 - Scope: $Q(y, w)$
 - Bound variable: w
- **Free variables:** x, y, z .
 - Note: The variables y and z are free in $P(x, y, z)$. The x in $Q(x, u)$ is free because it is outside the scope of $\forall x$. The y in $Q(y, w)$ is also free.

Q4 Solution

1. Define Predicates

Let the domain be all people:

- $M(x)$: x is a mathematician.
- $L(x)$: x is a logical thinker.
- $C(x)$: x is creative.

2. Translate Statements into Predicate Logic

- Premise 1: $\forall x (M(x) \rightarrow L(x))$
- Premise 2: $\exists x (M(x) \wedge C(x))$
- Conclusion: $\exists x (L(x) \wedge C(x))$

3. Formal Deduction Proof

Step	Statement	Reason
1	$\exists x (M(x) \wedge C(x))$	Premise (2)
2	$M(a) \wedge C(a)$	Existential Instantiation (from Step 1)
3	$M(a)$	Simplification (from Step 2)
4	$C(a)$	Simplification (from Step 2)
5	$\forall x (M(x) \rightarrow L(x))$	Premise (1)
6	$M(a) \rightarrow L(a)$	Universal Instantiation (from Step 5)
7	$L(a)$	Modus Ponens (from Steps 3 and 6)
8	$L(a) \wedge C(a)$	Conjunction (from Steps 7 and 4)
9	$\exists x (L(x) \wedge C(x))$	Existential Generalization (from Step 8)

Q5 Solution

1. Define Predicates

Let the domain of discourse be all birds. We define the following predicates:

- $S(x)$: x sings beautifully.
- $C(x)$: x is brightly colored.
- $L(x)$: x is a large bird.
- $H(x)$: x primarily feeds on honey.

2. Translate to Predicate Logic

Premises:

1. All birds that sing beautifully are brightly colored:

$$\forall x (S(x) \rightarrow C(x))$$

2. No large birds primarily feed on honey:

$$\forall x (L(x) \rightarrow \neg H(x))$$

3. Birds that do not primarily feed on honey have dull colors:

$$\forall x (\neg H(x) \rightarrow \neg C(x))$$

Conclusion:

- No birds that sing beautifully are large birds:

$$\forall x (S(x) \rightarrow \neg L(x))$$

3. Formal Deduction (Proof)

Let a represent an arbitrary bird in our domain.

Step	Statement	Reason
1	$\forall x (S(x) \rightarrow C(x))$	Premise (1)
2	$\forall x (L(x) \rightarrow \neg H(x))$	Premise (2)
3	$\forall x (\neg H(x) \rightarrow \neg C(x))$	Premise (3)
4	$S(a) \rightarrow C(a)$	Universal Instantiation (from Step 1)
5	$L(a) \rightarrow \neg H(a)$	Universal Instantiation (from Step 2)
6	$\neg H(a) \rightarrow \neg C(a)$	Universal Instantiation (from Step 3)
7	$L(a) \rightarrow \neg C(a)$	Hypothetical Syllogism (from Steps 5 and 6)
8	$C(a) \rightarrow \neg L(a)$	Contrapositive (from Step 7)
9	$S(a) \rightarrow \neg L(a)$	Hypothetical Syllogism (from Steps 4 and 8)
10	$\forall x (S(x) \rightarrow \neg L(x))$	Universal Generalization (from Step 9) ■